Lecture 4 v3:
Running a Regression in Excel

25117 - Econometrics

Universitat Pompeu Fabra

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From 100 schools to a regression result

1. Open and check a dataset with 100 synthetic schools.
2. Draw a scatterplot and estimate a simple regression in Excel.
3. Read coefficients, fit measures, standard errors and confidence intervals.
4. Calculate fitted values and residuals yourself.
5. Introduce family income, Z, to prepare for Lecture 5.

What you need

Desktop Excel for Windows or Mac and `school_regression.xlsx`. Use the `Data` tab first; `Worked` contains the formula-based answers.

Our question: do schools with smaller classes score higher?

- One observation is one school, not one pupil: $n = 100$ schools.
- Y: the school's average test score, measured in points.
- X: average class size, measured in students per class. A school average can be a decimal.
- Later, Z: average annual family income at the school, measured in **thousands of euros**.

Interpretation

These are simulated data designed to resemble an observational school study. A difference in scores or an estimated slope does not, by itself, establish causality.

The dataset: one row per school, one column per variable

A: school ID	B: score y_i	C: class size x_i	D: income z_i
1	610.8186	27.8	23.39
2	643.7577	19.7	33.48
3	658.5047	17.6	35.09
4	663.7172	16.3	36.68
5	642.3621	24.6	27.42
6	660.9807	21.0	46.49
⋮	⋮	⋮	⋮
100	The workbook contains all 100 observations.		

- Row 1 contains labels; rows 2–101 contain the observations.
- Start with columns B and C. Keep column D for the end of the lecture.
- The numbers are fixed: everyone should obtain the same answers.

The model we will estimate

$$Y_i = \beta_0 + \beta_1 X_i + U_i.$$

- β_0: intercept; β_1: slope; U_i: factors not included in X_i.
- Excel chooses estimates that minimize the sum of squared residuals:

$$\min_{b_0, b_1} \sum_{i=1}^{100} (y_i - b_0 - b_1 x_i)^2.$$

- After estimation, $\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$ and $\hat{u}_i = y_i - \hat{y}_i$.
- We first describe the association between class size and scores. A causal interpretation requires additional assumptions.

Step 1. Open the file and check its structure

1. Open `school_regression.xlsx` in desktop Excel. Save a working copy.
2. Select `Data`. Check the labels in A1:D1 and the last school in row 101.
3. Click B2: the formula bar should show a number, not text or a random formula.
4. Keep each school's values on the same row. If sorting, select the entire table.

Excel language and regional settings

The slides use English menu and function names, decimal points, and commas between function arguments. A localized Excel installation may use translated names and semicolons. The supplied workbook avoids CSV import issues.

Step 2. Check the number of observations and basic statistics

In an empty cell, type a formula beginning with =, then press Enter.

Check	Excel formula	Expected value
Number of scores	`=COUNT(Data!B2:B101)`	100
Number of class sizes	`=COUNT(Data!C2:C101)`	100
Sample mean of Y	`=AVERAGE(Data!B2:B101)`	652.9419
Sample mean of X	`=AVERAGE(Data!C2:C101)`	21.7280
Sample SD of Y	`=STDEV.S(Data!B2:B101)`	20.8846
Minimum class size	`=MIN(Data!C2:C101)`	16.1
Maximum class size	`=MAX(Data!C2:C101)`	27.9

If a count is below 100, check for missing values or numbers stored as text.

Step 3. Create an XY scatterplot

1. Select `Data!B1:C101`. Choose **Insert** → **Scatter (X, Y)** → **markers only**.
2. Check the axes: class size must be horizontal; score must be vertical.
3. To set them explicitly: select the chart, then **Chart Design** → **Select Data**. Edit the series (remove any extra series).

Series settings

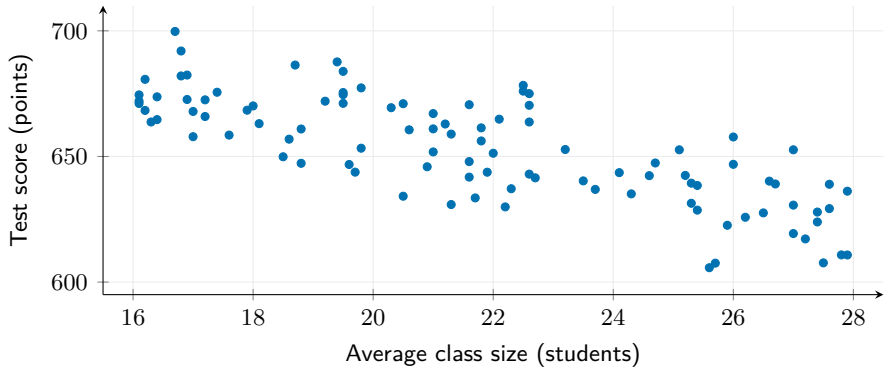
Series name: `Test score`

Series X values: `=Data!$C$2:$C$101`

Series Y values: `=Data!$B$2:$B$101`

Add axis titles: **Average class size (students)** and **Test score (points)**.

What the scatterplot should look like



Each dot is one school. The association is negative, with substantial variation around it.

Step 4. Add a fitted line to the chart

1. Select a point in the chart.
2. Right-click and choose **Add Trendline**.
3. Select **Linear**.
4. Check **Display Equation on chart** and **Display R-squared value on chart**.

Do not force the intercept to zero.

Expected equation

$$\hat{y}_i = 749.0587 - 4.4236 x_i$$

$$R^2 = 0.5977.$$

The chart may round its labels. Use regression output or worksheet formulas for calculations.

A trendline is a quick check. The Regression tool also reports standard errors and confidence intervals.

Step 5. Enable the Analysis ToolPak

Windows

1. **File** → **Options** → **Add-ins**.
2. At the bottom, choose **Manage: Excel Add-ins** → **Go**.
3. Check **Analysis ToolPak**; click **OK**.

Mac

1. **Tools** → **Excel Add-ins**.
2. Check **Analysis ToolPak**; click **OK**.
3. Restart Excel if necessary.

Now look for **Data** → **Data Analysis**. Select **Regression**.

Use desktop Excel: the standard ToolPak Regression command is unavailable in Excel for the web.

Step 6. Fill in the Regression dialog

Field or option	What to enter or select
Input Y Range	`Data!$B$1:$B$101`
Input X Range	`Data!$C$1:$C$101`
Labels	Check: our selected ranges include row 1
Constant is Zero	**Leave unchecked**
Confidence Level	95%
New Worksheet Ply	Select; enter `OLS_simple`
Residuals	Check
Residual Plots	Check

Click **OK**. Excel creates a new sheet containing the regression results.

Do not include school ID or income in this first X range.

Step 7. Read the regression summary

Regression Statistics	Value
Multiple R	0.7731
R Square	0.5977
Adjusted R Square	0.5936
Standard Error	13.3138
Observations	100

- **R Square**: the fitted line accounts for about 59.77% of the sample variation in scores.
- **Standard Error**: the SER, measured in score points. It is not the standard error of the slope.
- **Multiple R**: here, $\sqrt{R^2}$. It is nonnegative, even though the slope is negative.
- **Observations**: check that Excel used all 100 schools.

The ANOVA block decomposes variation. For this exercise, focus first on the summary and coefficient rows.

Step 8. Find the coefficients and their uncertainty

	Coefficient	Std. Error	t Stat	P-value	Lower 95%	Upper 95%
Intercept	749.0587	8.0760	92.7514	2.75×10^{-97}	733.0322	765.0853
X_class_size	-4.4236	0.3666	-12.0667	4.33×10^{-21}	-5.1511	-3.6961

- Read the **X_class_size** row for the estimated slope and its standard error.
- Excel's **P-value** tests a coefficient equal to zero against a two-sided alternative.
- Scientific notation such as `4.33E-21` means 4.33×10^{-21}, not zero.
- These are conventional, homoskedastic standard errors. ToolPak does not automatically report heteroskedasticity-robust standard errors.

Translate the estimated line into a sentence

$$\widehat{\text{score}} = 749.0587 - 4.4236\,\text{class size}.$$

- A school with an average class size larger by one student is predicted to score about **4.42 points lower**.
- A difference of five students corresponds to $5(-4.4236) = -22.12$ predicted score points.
- The intercept is the predicted score at a class size of zero. That is outside our data range, so it has no useful school interpretation here.
- The fitted association does not show what would happen if a school changed its class size. Family background may differ across schools.

Step 9. Reproduce the coefficients with Excel formulas

Create a new sheet named `Calculations`. Enter these labels and formulas:

Cell in column A	Formula in column B	Expected value
A2: Intercept	`=INTERCEPT(Data!B2:B101,Data!C2:C101)`	749.0587
A3: Slope	`=SLOPE(Data!B2:B101,Data!C2:C101)`	-4.4236
A4: n	`=COUNT(Data!B2:B101)`	100
A5: R squared	`=RSQ(Data!B2:B101,Data!C2:C101)`	0.5977

- In each function, enter the Y range first and the X range second.
- Show four decimals for readability, but keep the full underlying values.
- Unlike ToolPak output, these formulas update if the data change. Rerun ToolPak after changing data.

Step 10. Calculate fitted values and residuals

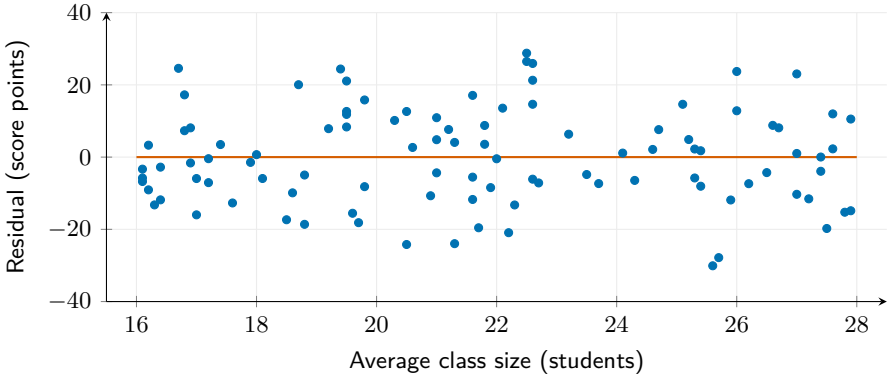
Return to `Data`. Add three column labels and enter formulas in row 2:

New column	Formula in the first data row
E1: Y_fitted	E2: `=Calculations!$B$2+Calculations!$B$3*C2`
F1: Residual	F2: `=B2-E2`
G1: Residual_squared	G2: `=F2^2`

1. Select E2:G101 and use **Home** $\rightarrow$ **Fill** $\rightarrow$ **Down**.
2. The `$` signs keep the coefficient cells fixed; `C2` changes to `C3`, and so on.
3. Check school 1: $\hat{y}_1 \approx 626.0816$ and $\hat{u}_1 \approx -15.2630$.

Meaning: school 1 scored about 15.26 points below its fitted value.

Step 11. Inspect the residual plot



Use X values C2:C101 and Y values F2:F101 in an XY scatterplot. Look for curvature, unusually large residuals, or changing spread; this plot alone cannot establish causality.

Step 12. Calculate SSR, RMSE and SER

On `Calculations`, continue the table:

Label	Formula	Expected value
A6: SSR	B6: `=SUM(Data!G2:G101)`	17371.2137
A7: RMSE	B7: `=SQRT(B6/B4)`	13.1800
A8: SER	B8: `=SQRT(B6/(B4-2))`	13.3138

$$SSR = \sum_{i=1}^{100} \hat{u}_i^2, \text{ and } RMSE = \sqrt{\frac{SSR}{100}}.$$

$$SER = \sqrt{\frac{SSR}{100-2}}.$$

Two estimated coefficients leave $100 - 2 = 98$ residual degrees of freedom. Excel's summary **Standard Error** is SER, not RMSE.

Step 13. Test whether the slope is zero

$$H_0 : \beta_1 = 0, \text{ and } H_1 : \beta_1 \neq 0.$$

On `Calculations`, enter:

Label	Formula	Value
A9: SE of slope	B9: `=B8/SQRT(DEVSQ(Data!C2:C101))`	0.3666
A10: t statistic	B10: `=B3/B9`	-12.0667

$$t_{\text{obs}} = \frac{-4.4236 - 0}{0.3666} \approx -12.07.$$

- Using the normal approximation already covered: $|t_{\text{obs}}| > 1.96$, so reject H_0 at 5%.
- Excel reports a two-sided p-value of about 4.33×10^{-21} using a t reference distribution with 98 degrees of freedom. The decision is the same.

Step 14. Build and interpret a confidence interval

Label	Formula	Value
A11: Normal critical value	B11: `=1.96`	1.96
A12: Lower endpoint	B12: `=B3-B11*B9`	-5.1422
A13: Upper endpoint	B13: `=B3+B11*B9`	-3.7051

$$\hat{\beta}_1 \pm 1.96\, SE(\hat{\beta}_1) \approx [-5.14, -3.71].$$

- 1.96 leaves 95% of a standard normal distribution between -1.96 and 1.96.
- Excel's **Lower 95%** and **Upper 95%** use 1.9845 from its t reference distribution: $[-5.1511, -3.6961]$.
- Both intervals exclude zero. Excluding zero is equivalent to rejecting the corresponding two-sided 5% test.

Write a short regression report

Example

Across 100 synthetic schools, average class size is negatively associated with average test scores. An additional student per class is associated with approximately 4.42 fewer score points (standard error 0.37; $p < 0.001$). The fitted line accounts for 59.77% of the sample variation in scores, and the SER is 13.31 points.

- Include the sample size, units, coefficient and uncertainty.
- Distinguish statistical significance from the size of the association.
- State the limitation: schools differ in family background and other factors, so this regression alone does not identify a causal class-size effect.

Now introduce Z: family income

- Column D contains a new explanatory variable for the same 100 schools: `Z_income_kEUR`.
- A value of 40 means an average annual family income of €40,000.
- Income may affect learning resources and therefore test scores. It may also be related to class size.
- Every income value must remain matched to the same school ID, score and class size.

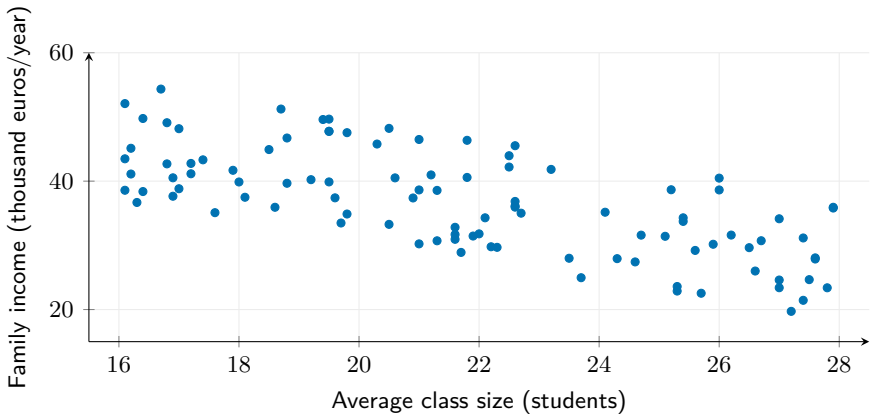
Check it in Excel

`=AVERAGE(Data!D2:D101)` gives 36.6975.

`=CORREL(Data!C2:C101,Data!D2:D101)` gives -0.6976.

The dataset already includes Z, so the transition to Lecture 5 uses exactly the same observations.

Schools with smaller classes also have higher family income



Recreate this graph with C2:C101 on the horizontal axis and D2:D101 on the vertical axis. Sample correlation: -0.6976.

Why could omitting income change the slope?

1. Income is a determinant of scores in our simulated example.
2. Income is negatively correlated with class size.
3. A comparison of large and small classes therefore also compares schools with different family incomes.

Connection to Lecture 5: omitted-variable bias

Both conditions matter: Z helps determine Y, and Z is correlated with X. The simple slope combines the class-size relationship with the income differences associated with class size.

With a positive income effect and a negative income–class-size relationship, we expect the simple slope to be more negative than the slope controlling for income.

Prepare the multiple regression for Lecture 5

$$Y_i = \alpha_0 + \alpha_1 X_i + \alpha_2 Z_i + \varepsilon_i.$$

1. Return to **Data** → **Data Analysis** → **Regression**.
2. Keep Input Y Range: `Data!$B$1:$B$101`.
3. Change Input X Range to `Data!$C$1:$D$101`: two adjacent columns, class size *and* income.
4. Check **Labels**, keep the intercept, and use a new output sheet named `OLS_multiple`.

The new question

How are class size and scores related **holding average family income constant**? We use different symbols, α, because these coefficients belong to a different model.

Preview: what changes when we include income?

	Simple regression	Multiple regression
Intercept	749.0587	646.2105
Class-size coefficient	−4.4236	−2.1867
Income coefficient	Not included	1.4781
R^2	0.5977	0.7590
SER	13.3138	10.3588
Observations	100	100

- Holding income constant, one additional student per class is associated with about 2.19 fewer score points.
- Holding class size constant, an additional €1,000 of income is associated with about 1.48 more score points.
- Adding a regressor cannot lower in-sample R^2. A higher R^2 alone does not establish causality.

Your turn: six checks before leaving Excel

1. Predict the score for a school with an average class size of 20 using the simple regression.
2. What happens to the fitted score if class size rises from 20 to 25?
3. Calculate `=SUM(Data!F2:F101)`. Why is it approximately zero?
4. Which number is the slope's standard error: 13.3138 or 0.3666?
5. Why does the normal-approximation confidence interval exclude zero?
6. Compare the two class-size coefficients. What does adding income change about the comparison?

Answers and a troubleshooting checklist

Answers

1. $749.0587 - 4.4236(20) \approx 660.5860$.
2. About 22.1182 points lower.
3. OLS with an intercept makes residuals sum to zero, up to rounding.
4. 0.3666; 13.3138 is the SER.
5. The estimate is more than 1.96 standard errors from zero.
6. We now compare class sizes holding income constant.

If your output differs

- Confirm Y is column B and X is column C.
- Use rows 1–101 with **Labels** checked, or 2–101 without it.
- Leave **Constant is Zero** unchecked.
- Exclude IDs and missing cells.
- Keep full precision; rerun ToolPak after changing data.

Behind the data: how the 100 schools were created

Independent draws across schools, with fixed seed 251174:

$$
\begin{aligned}
X_i &\sim \text{Uniform}(16, 28), \\
V_i &\sim \text{Uniform}(-10, 10), \text{ and } \varepsilon_i \sim N(0, 10^2), \\
Z_i &= 70 - 1.5X_i + V_i, \\
U_i &= 1.5(Z_i - 40) + \varepsilon_i, \\
Y_i &= 700 - 2X_i + U_i = 640 - 2X_i + 1.5Z_i + \varepsilon_i.
\end{aligned}
$$

- X_i, V_i and ε_i are drawn independently. Stored class sizes have one decimal, incomes two, and scores four; see the generation script for the rounding order.
- This has the form $Y = b_0 + b_1 X + U$, with $b_0 = 700$ and $b_1 = -2$; the omitted income component makes $E(U \mid X) \neq 0$.
- Therefore -2 is the class-size effect holding income constant in this simulation, not the slope of the simple conditional mean. Regressing on X alone need not recover it.

Files and Excel help

- `data/lecture4v3/school_regression.xlsx`: fixed data and a worked formula sheet.
- `data/lecture4v3/school_data.csv`: the same 100 observations in a portable format.
- `data/lecture4v3/generate_data.py`: reproduces the synthetic observations and independently checks the regression results.
- Microsoft Support: Load the Analysis ToolPak in Excel.
- Microsoft Support: Use the Analysis ToolPak to perform complex data analysis.
- Microsoft Support: Perform a regression analysis.

The figures and benchmark results use the saved dataset. Menu names can vary by Excel version and language.